

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 23

1

Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.3, 16.4 (fundamental theorem of line integrals and Green's theorem)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

2



ALVARO: Start the recording!



3

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Parametric Surfaces

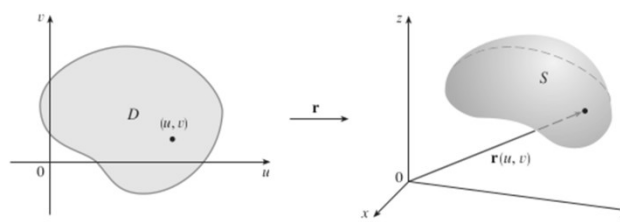


4

About Parametrized Surfaces

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

A parametric surface



5

Example: Identify and sketch the surface with vector equation

$$\mathbf{r}(u, v) = (1, u, u + v) = 1 \cdot \mathbf{i} + u \cdot \mathbf{j} + (u + v) \cdot \mathbf{k}.$$

6

[Extra Space]

Example: Identify and sketch the surface with vector equation

$$r(u, v) = (1, u, u + v) = 1 \cdot i + u \cdot j + (u + v) \cdot k.$$



7

Example: Identify and sketch the surface with vector equation

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8

[Extra Space]

Example: Identify and sketch the surface with vector equation

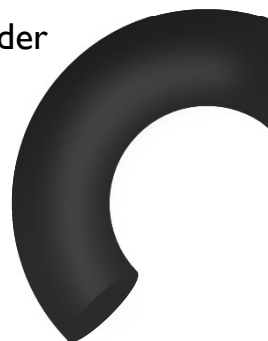
$$r(u, v) = (u, u, u + v) = u \cdot i + u \cdot j + (u + v) \cdot k.$$



9

Example: Find a parametric equation for the segment of a cylinder

$$x^2 + y^2 = 4, \quad 0 \leq z \leq 1.$$

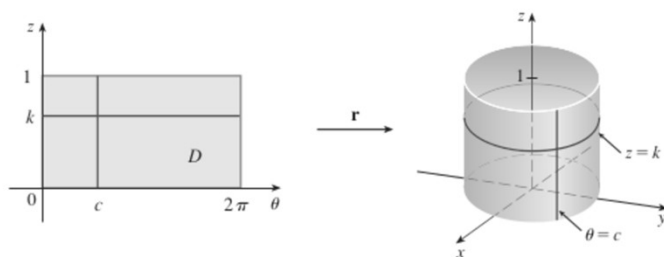


10

[Extra Space]

Example: Find a parametric equation for the segment of a cylinder

$$x^2 + y^2 = 4, \quad 0 \leq z \leq 1.$$



11

Example: Identify and sketch the surface with vector equation

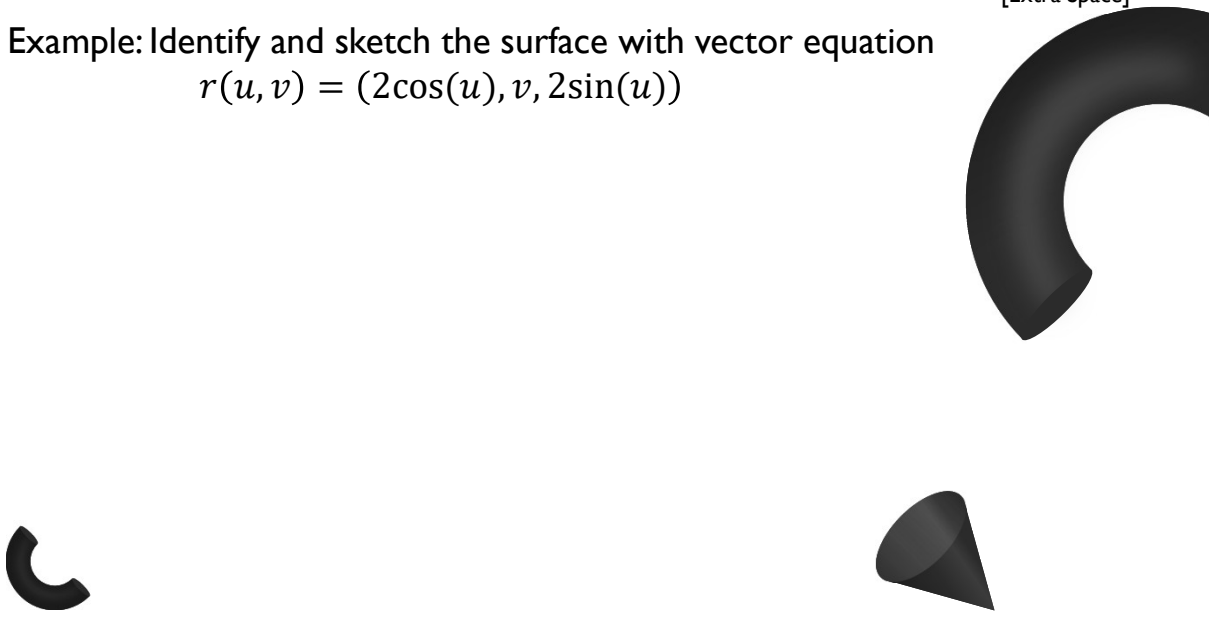
$$\mathbf{r}(u, v) = (2\cos(u), v, 2\sin(u))$$



12

[Extra Space]

Example: Identify and sketch the surface with vector equation

$$r(u, v) = (2\cos(u), v, 2\sin(u))$$


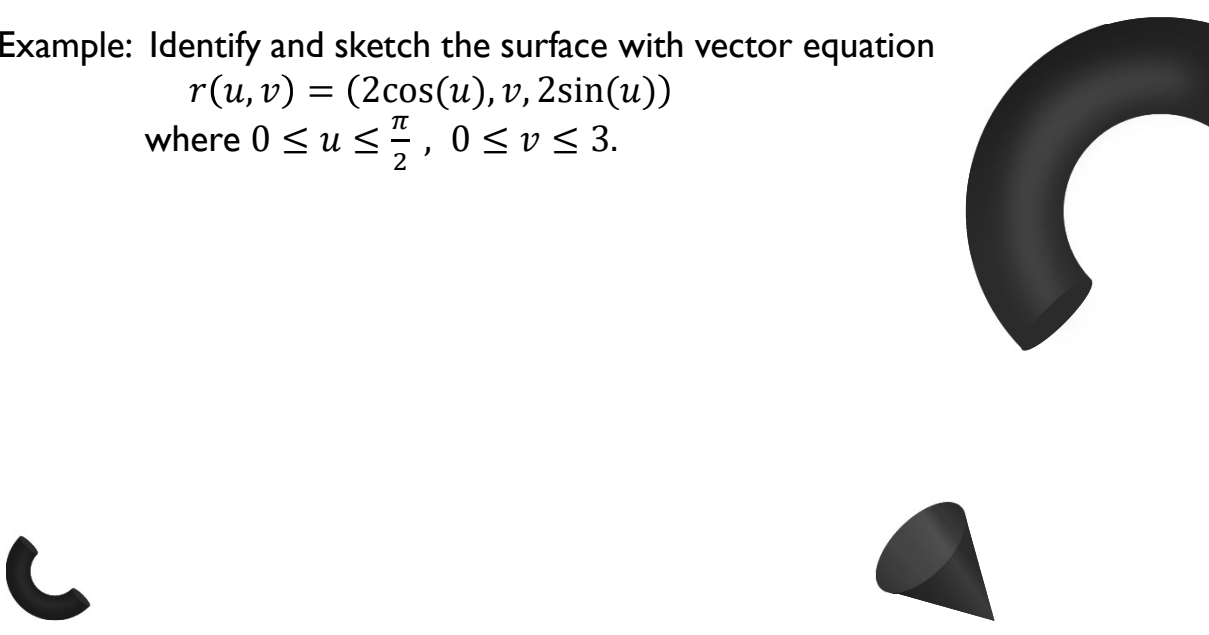
The image displays three 3D surface plots. The top right plot is a large, thick, curved tube that follows a quarter-circle path in the xy-plane, extending along the z-axis. The bottom left plot is a smaller version of this curved tube. The bottom right plot is a cone with its vertex at the origin, opening upwards.

13

Example: Identify and sketch the surface with vector equation

$$r(u, v) = (2\cos(u), v, 2\sin(u))$$

where $0 \leq u \leq \frac{\pi}{2}$, $0 \leq v \leq 3$.



The image displays three 3D surface plots. The top right plot is a large, thick, curved tube that follows a quarter-circle path in the xy-plane, extending along the z-axis. The bottom left plot is a smaller version of this curved tube. The bottom right plot is a cone with its vertex at the origin, opening upwards.

14

[Extra Space]

Example: Identify and sketch the surface with vector equation

$$r(u, v) = (2\cos(u), v, 2\sin(u))$$

where $0 \leq u \leq \frac{\pi}{2}$, $0 \leq v \leq 3$.



15

Example: Find a parametric equation for the elliptic paraboloid

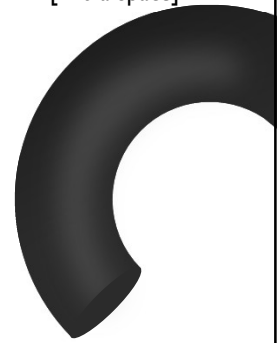
$$z = x^2 + 2y^2.$$



16

Example: Find a parametric equation for the elliptic paraboloid
 $z = x^2 + 2y^2$.

[Extra Space]



17

Parametric surfaces can be complicated!

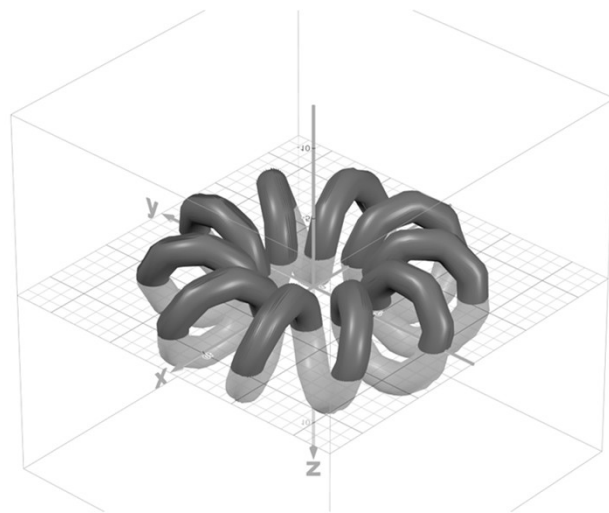
$(X(u,v), Y(u,v), Z(u,v))$

$0 \leq u \leq 1$
 $0 \leq v \leq 1$

$X(u,v) = \cos(rv) \cos(tu) + (\cos(10rv)(\sin(tu) + 3) + 7)(-\sin(rv))$

$Y(u,v) = \sin(rv) \cos(tu) + (\cos(10rv)(\sin(tu) + 3) + 7)(\cos(rv))$

$Z(u,v) = (\sin(10rv))(\sin(tu) + 3)$



18

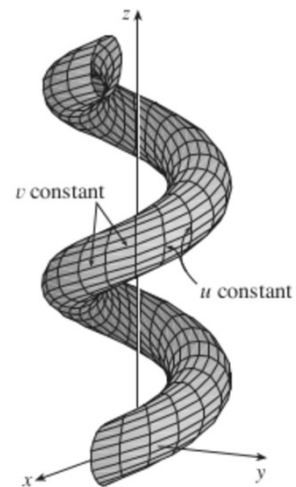
Grid curves on parametric surfaces



19

Example: Sketch the “grid curves” with constant v

$$r(u, v) = ((2 + \sin(v)) \cos(u), (2 + \sin(v)) \sin(u), u + \cos(v))$$

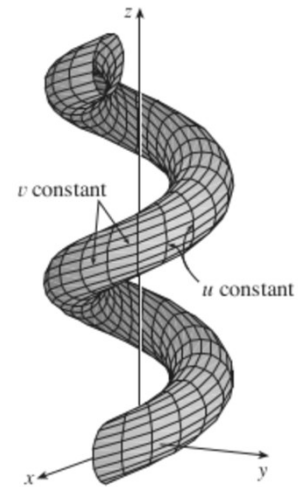


20

[Extra Space]

Example: Sketch the “grid curves” with constant v

$$r(u, v) = ((2 + \sin(v)) \cos(u), (2 + \sin(v)) \sin(u), u + \cos(v))$$



21

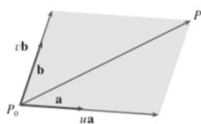
Example: Find a vector equation for the plane that passes through a point P and that contains two non-parallel vectors a and b .

22

[Extra Space]

Example: Find a vector equation for the plane that passes through a point P and that contains two non-parallel vectors $\mathbf{a}$ and $\mathbf{b}$.

Figure 6



If we write $\mathbf{r} = \langle x, y, z \rangle$, $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$, $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then we can write the parametric equations of the plane through the point (x_0, y_0, z_0) as follows:

$$x = x_0 + ua_1 + vb_1 \quad y = y_0 + ua_2 + vb_2 \quad z = z_0 + ua_3 + vb_3$$

23

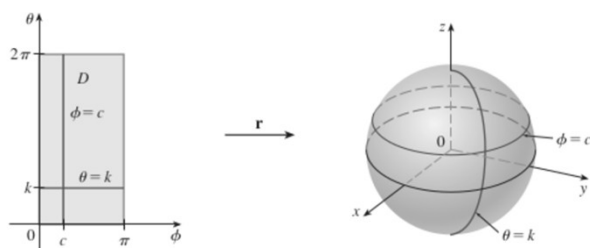
Example: Find a parametric equation for a sphere centered at $(0,0,0)$ of radius R .

24

Example: Find a parametric equation for a sphere centered at $(0,0,0)$ of radius R .

[Extra Space]

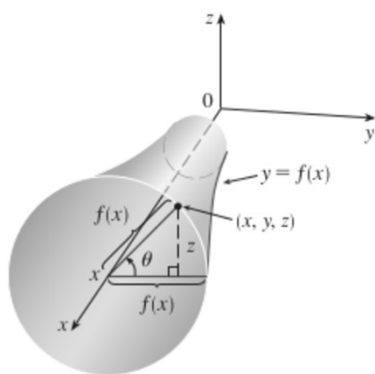
$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$



25

Surfaces of Revolution

$$x = x \quad y = f(x) \cos \theta \quad z = f(x) \sin \theta$$



26

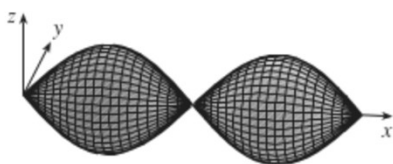
Example: Find a parametric equation for the surface that is obtained by rotating the curve $y = \sin(x)$, $0 \leq x \leq 2\pi$, about the x -axis.



27

[Extra Space]

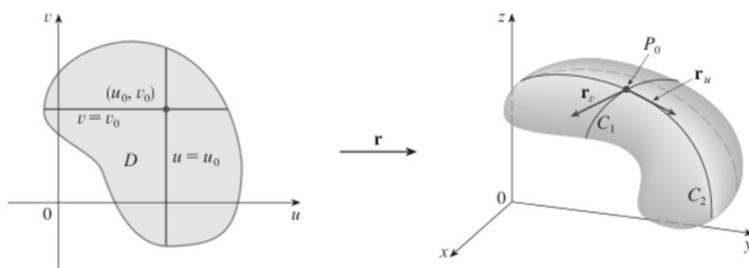
Example: Find a parametric equation for the surface that is obtained by rotating the curve $y = \sin(x)$, $0 \leq x \leq 2\pi$, about the x -axis.



28

Tangent Planes

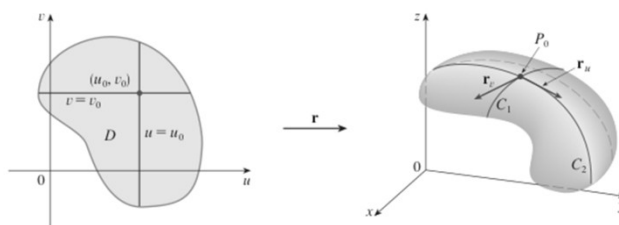
$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$



29

Tangent Planes

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

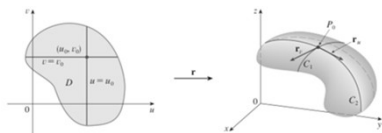


$$\mathbf{r}_u = \frac{\partial x}{\partial u}(u_0, v_0) \mathbf{i} + \frac{\partial y}{\partial u}(u_0, v_0) \mathbf{j} + \frac{\partial z}{\partial u}(u_0, v_0) \mathbf{k}$$

$$\mathbf{r}_v = \frac{\partial x}{\partial v}(u_0, v_0) \mathbf{i} + \frac{\partial y}{\partial v}(u_0, v_0) \mathbf{j} + \frac{\partial z}{\partial v}(u_0, v_0) \mathbf{k}$$

30

Tangent Planes



$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

$$\mathbf{r}_u = \frac{\partial x}{\partial u}(u_0, v_0) \mathbf{i} + \frac{\partial y}{\partial u}(u_0, v_0) \mathbf{j} + \frac{\partial z}{\partial u}(u_0, v_0) \mathbf{k}$$

$$\mathbf{r}_v = \frac{\partial x}{\partial v}(u_0, v_0) \mathbf{i} + \frac{\partial y}{\partial v}(u_0, v_0) \mathbf{j} + \frac{\partial z}{\partial v}(u_0, v_0) \mathbf{k}$$

If $\mathbf{r}_u \times \mathbf{r}_v$ is not $\mathbf{0}$, then the surface S is called **smooth** (it has no “corners”). For a smooth surface, the **tangent plane** is the plane that contains the tangent vectors $\mathbf{r}_u$ and $\mathbf{r}_v$, and the vector $\mathbf{r}_u \times \mathbf{r}_v$ is a normal vector to the tangent plane.

31

Example: Find the equation of the tangent plane to the surface with parametric equations $x = u^2, y = v^2, z = u + 2v$ at the point $P=(1,1,3)$.

32

[Extra Space]

Example: Find the equation of the tangent plane to the surface with parametric equations $x = u^2, y = v^2, z = u + 2v$ at the point $P=(1,1,3)$.

33

Questions?

34



ALVARO: Start the recording!



35

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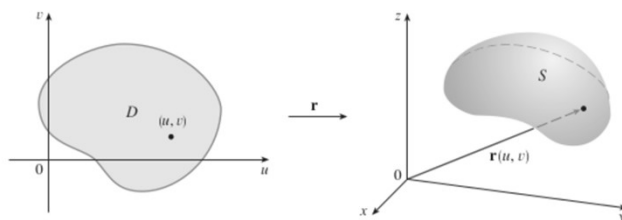
Surface Area

36

About Parametrized Surfaces and their Area

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

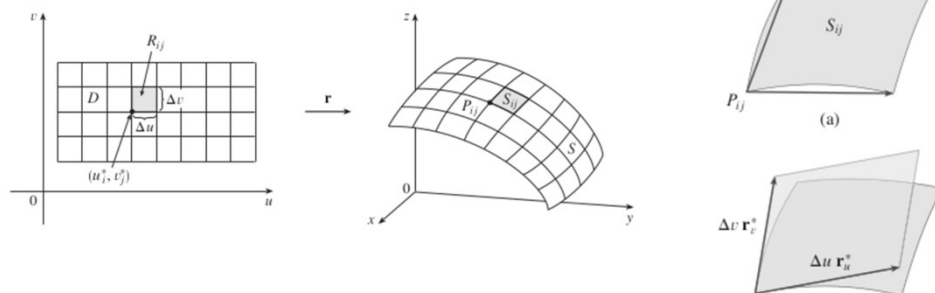
A parametric surface



37

About Parametrized Surfaces and their Area

The image of the subrectangle R_{ij} is the patch S_{ij} .



$$|(\Delta u \mathbf{r}_u^*) \times (\Delta v \mathbf{r}_v^*)| = |\mathbf{r}_u^* \times \mathbf{r}_v^*| \Delta u \Delta v$$

$$\sum_{i=1}^m \sum_{j=1}^n |\mathbf{r}_u^* \times \mathbf{r}_v^*| \Delta u \Delta v$$

38

About Parametrized Surfaces and their Area

If a smooth parametric surface S is given by the equation

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k} \quad (u, v) \in D$$

and S is covered just once as (u, v) ranges throughout the parameter domain D , then the **surface area** of S is

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| \, dA$$

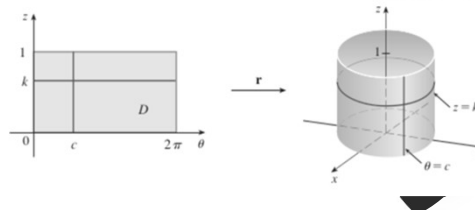
where

$$\mathbf{r}_u = \frac{\partial x}{\partial u} \mathbf{i} + \frac{\partial y}{\partial u} \mathbf{j} + \frac{\partial z}{\partial u} \mathbf{k} \quad \mathbf{r}_v = \frac{\partial x}{\partial v} \mathbf{i} + \frac{\partial y}{\partial v} \mathbf{j} + \frac{\partial z}{\partial v} \mathbf{k}$$

39

Example: Compute the surface area of the segment of a cylinder

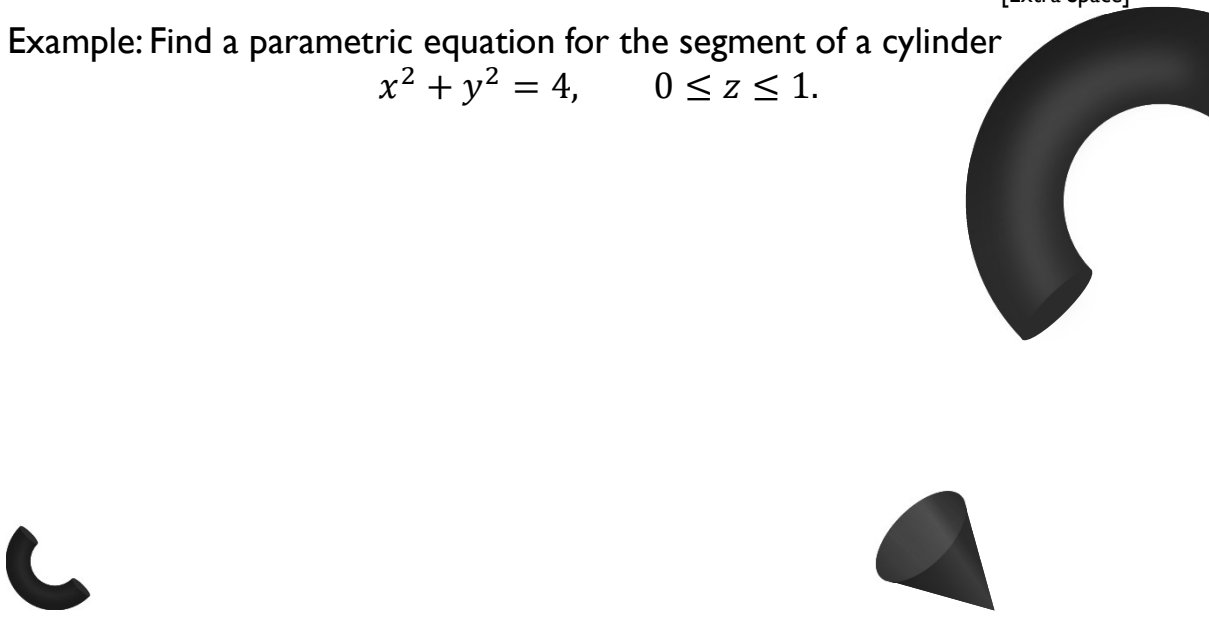
$$x^2 + y^2 = 4, \quad 0 \leq z \leq 1.$$



40

[Extra Space]

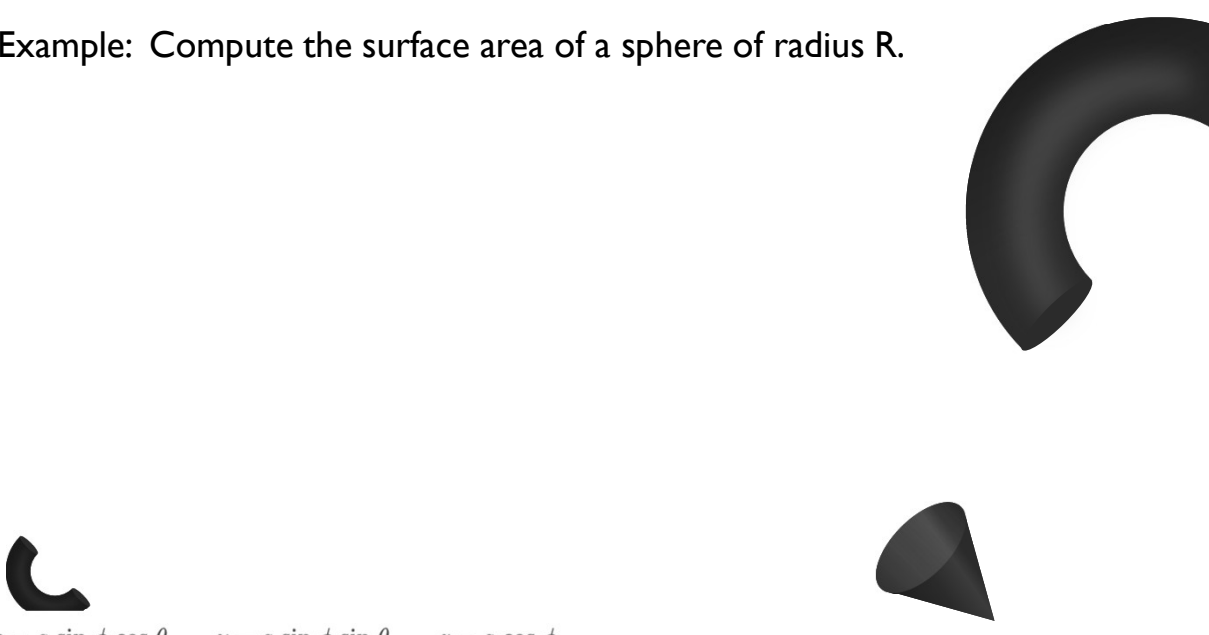
Example: Find a parametric equation for the segment of a cylinder
 $x^2 + y^2 = 4, \quad 0 \leq z \leq 1.$



The image contains three 3D models. In the top right is a large, dark gray curved surface representing a segment of a cylinder. In the bottom left is a smaller, similar curved surface. In the bottom right is a dark gray cone pointing downwards.

41

Example: Compute the surface area of a sphere of radius R.



The image contains three 3D models. In the top right is a large, dark gray curved surface representing a segment of a cylinder. In the bottom left is a smaller, similar curved surface. In the bottom right is a dark gray cone pointing downwards.

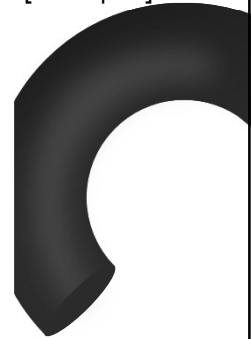
$x = a \sin \phi \cos \theta$ $y = a \sin \phi \sin \theta$ $z = a \cos \phi$

42

Example: Compute the surface area of a sphere of radius R.

[Extra Space]

$$\begin{aligned}\mathbf{r}_\phi \times \mathbf{r}_\theta &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial x}{\partial \phi} & \frac{\partial y}{\partial \phi} & \frac{\partial z}{\partial \phi} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} & \frac{\partial z}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a \cos \phi \cos \theta & a \cos \phi \sin \theta & -a \sin \phi \\ -a \sin \phi \sin \theta & a \sin \phi \cos \theta & 0 \end{vmatrix} \\ &= a^2 \sin^2 \phi \cos \theta \mathbf{i} + a^2 \sin^2 \phi \sin \theta \mathbf{j} + a^2 \sin \phi \cos \phi \mathbf{k}\end{aligned}$$



$$\begin{aligned}|\mathbf{r}_\phi \times \mathbf{r}_\theta| &= \sqrt{a^4 \sin^4 \phi \cos^2 \theta + a^4 \sin^4 \phi \sin^2 \theta + a^4 \sin^2 \phi \cos^2 \phi} \\ &= \sqrt{a^4 \sin^4 \phi + a^4 \sin^2 \phi \cos^2 \phi} = a^2 \sqrt{\sin^2 \phi} = a^2 \sin \phi\end{aligned}$$

43

Surfaces Area of a Graph $z = f(x,y)$



44

Surfaces Area of a Graph (and Length of a Graph)

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA$$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$



45

Example: Compute the surface area of the paraboloid
 $z = x^2 + y^2$ below the plane $z = 9$.



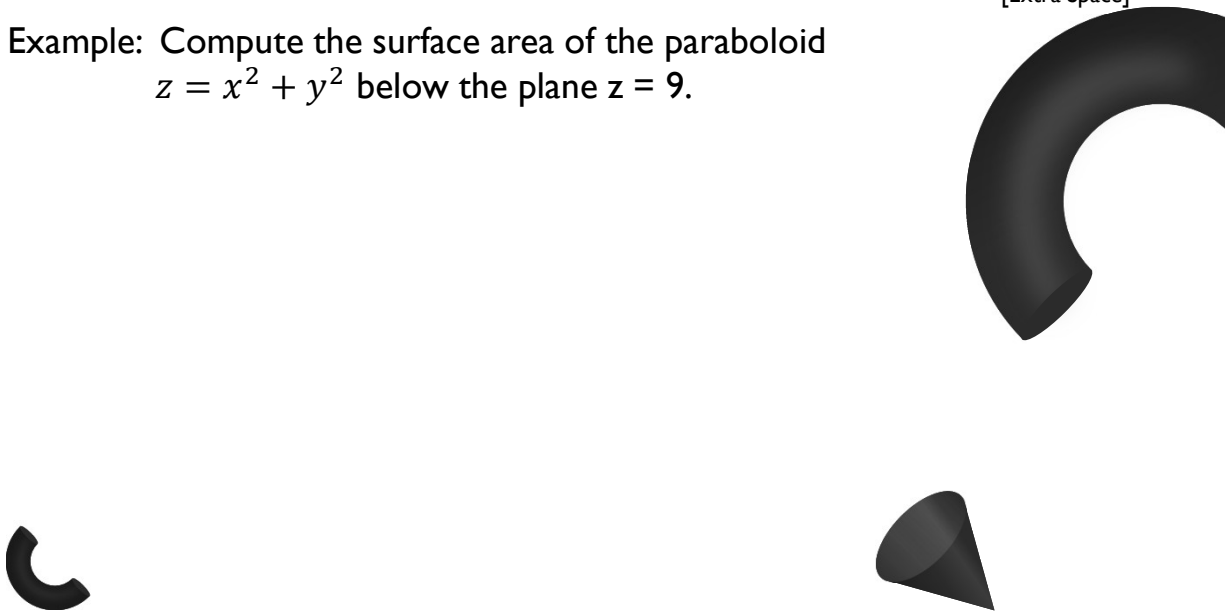
$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$



46

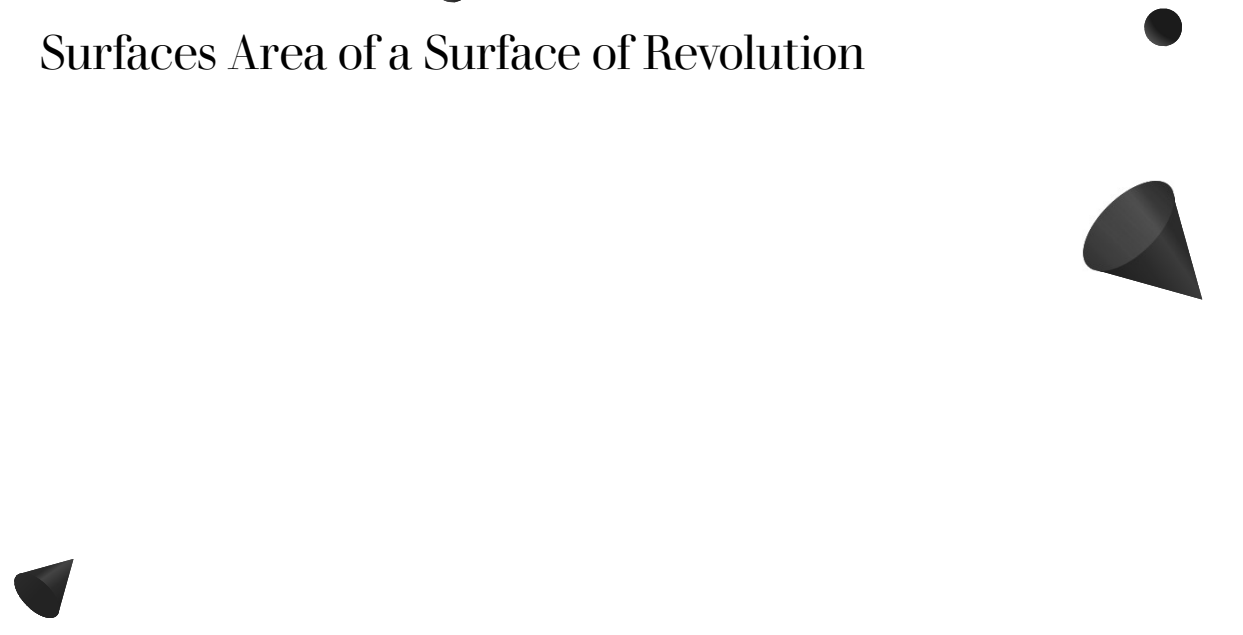
[Extra Space]

Example: Compute the surface area of the paraboloid
 $z = x^2 + y^2$ below the plane $z = 9$.



47

Surfaces Area of a Surface of Revolution



48

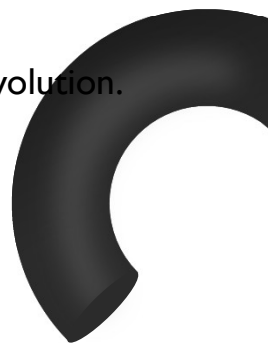
Surfaces Area of a Surface of Revolution

$$\begin{aligned}
 A &= \iint_D |\mathbf{r}_x \times \mathbf{r}_\theta| \, dA \\
 &= \int_0^{2\pi} \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \, dx \, d\theta \\
 &= 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \, dx
 \end{aligned}$$



49

Example: Compute the surface area of the paraboloid
 $z = x^2 + y^2$ below the plane $z = 9$, as a surface of revolution.



$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$

50

[Extra Space]

Example: Compute the surface area of the paraboloid
 $z = x^2 + y^2$ below the plane $z = 9$, as a surface of revolution.

51

Questions?

52

Thank you

Until next time.

